Sonu Pandit

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SKILLS

Languages: C, C++, Python

Tools/Platforms: MATLAB - Simulink, Proteus, Arduino IDE

Technical Skills: Power Systems, Electrical Machines, Power Electronics

Soft Skills: Problem-Solving, Teamwork, Analytical Thinking, Project Management, Adaptability

PROJECTS

AI-Based Power System Monitor and Protection — Apr' 2026

- Designed a smart protection unit to spot abnormal line conditions at an early stage, preventing damage to connected loads and keeping the electrical supply dependable.
- Trained an AI model on voltage and current readings from sensors to recognize healthy, overloaded, and faulty states, then programmed automatic circuit cut-off and a web dashboard for remote switching.
- Delivered detection within 1 second, a 100% trip success rate across tested scenarios, and about 500ms dashboard response, confirming consistent performance under live conditions.
- **Tech Stack:** Machine Learning, ESP32, ACS712, ZMPT101B, Relay, Wi-Fi

BI-Copter (A Two-Rotor Drone) — Jun'2025

- Constructed a compact aerial prototype to study balance and steady hovering through continuous feedback correction, laying the groundwork for unmanned aerial systems and autonomous flying platforms.
- Integrated propellers and orientation sensors with the motor drivers and tuned a PID loop to adjust the speed of each side instantly, keeping the frame level with 95% alignment accuracy across test sessions.
- Resolved vibration and mounting issues through repeated trials, enabling steady lift-off and flight at heights of 3–4.5 m and creating a solid base for further hardware upgrades.
- **Tech Included:** Arduino, BLDC Motors, ESC, PID Control

TRAINING

Tenughat Thermal Power Station (TTPS)|Link — 22 Jun 2026 – 28 Jul 2026

- Served as an Intern at the thermal facility, assisting the operations team with day-to-day electricity production and observing how a working plant runs from shift to shift.
- Explored the boiler, turbine, generator, transformer, and switchgear sections through site visits, working alongside technicians on routine tasks in each unit.
- Developed practical understanding of how generation equipment and protective systems operate together, building field skills relevant to plant engineering, grid management, and utility careers.

Electric Loco Shed Bokaro, Jharkhand|Link — 26 Aug 2024 - 25 Sept 2024

- Worked as a Trainee in the maintenance section of the railway workshop, supporting the upkeep of electric locomotives under experienced supervision and following shop-floor routines.
- Visited the areas handling traction motors, converters, circuit breakers, pantographs, braking systems, taking part in basic inspection and servicing across the shed.
- Acquired hands-on exposure to rolling-stock components and safe working habits, preparing for roles in railway electrification, overhaul, and drive-system engineering.

CERTIFICATES

- Cybersecurity Analyst Job Simulation | Tata Group, Forage |Link — Jul'2026
- MATLAB Onramp | MathWorks Training Services |Link — Jan'2026
- Programming in C | Infosys Springboard |Link — Jan'2026

ACHIEVEMENTS

- Published a patent titled "BI-COPTER" |Link — Dec'2025
- Secured 3rd prize in a Mathematics Quiz competition| Silli Polytechnic, Silli |Link — Dec'2022

EDUCATION

Lovely Professional University — Phagwara, Punjab
Bachelor of Technology in Electrical Engineering| CGPA: 8.53 — Aug' 25 – Present

Silli Polytechnic, Silli — Ranchi, Jharkhand
Diploma in Electrical Engineering| Percentage: 77.67 — Sep' 22 – June' 25

Krishna Sudarsan Central School (CCL) — Bokaro, Jharkhand
Matriculation| Percentage: 87.4 — Mar' 21 – Apr'22